Confidently Wrong: Measuring and Mitigating Calibration Risks in LLMs for African Languages[1]

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With

Apart Research

Abstract

Large language models (LLMs) are increasingly used in African languages, yet little is known about whether their confidence remains reliable. This creates a safety risk: models may become less accurate while remaining highly confident, increasing the likelihood of trusted misinformation. We evaluate four open-weight LLMs (**Qwen3-4B-Instruct**, **Gemma-4-12B**, **Aya-Expanse-8B**, and **AfroLlama-v1**) on African-language truthfulness and knowledge benchmarks (**Uhura-TruthfulQA** and **AfriMMLU**), measuring calibration using log-probability-based confidence estimates. We find that models become increasingly confident in incorrect answers as they move from English to African languages, creating a systematic overconfidence problem and a two- to three-fold increase in expected calibration error. Among three mitigation strategies, temperature scaling proves most effective, reducing calibration error to near-English levels without retraining. Our findings suggest that African-language deployment risks stem not only from lower accuracy, but also from models remaining confident when they are wrong. We show that simple post-hoc recalibration offers a practical deployment-time safety intervention.

1 Introduction

LLMs are entering high-stakes, information-seeking use in African languages such as health, legal, educational, and civic information, often where users cannot easily verify an answer. The safety failure we study is a mismatch between confidence and correctness. When models remain highly confident despite being wrong, they can mislead users into treating incorrect information as reliable, which is particularly dangerous in high-stakes settings such as health, education, and civic information. A calibrated model, whose confidence tracks its accuracy, can instead *abstain*, declining or deferring when uncertain [3]. If calibration holds in English but breaks in African languages, the populations least served by current models are most exposed to confident misinformation. We investigate whether models become increasingly overconfident in African languages, quantify the extent of this misalignment between confidence and correctness, and evaluate practical strategies for mitigating it.

Our main contributions are:

1. An open-weight, reproducible calibration audit of four LLMs on African-language truthfulness and knowledge benchmarks, using answer-text log-probabilities to obtain robust confidence estimates for instruction-tuned models.

[1]Research conducted at the Global South AI Safety Hackathon, June 2026.

2. Evidence that calibration degrades substantially from English to African languages, revealing systematic overconfidence and a growing mismatch between confidence and correctness across models and tasks.
3. An evaluation of practical mitigation strategies, including temperature scaling, abstention, and few-shot prompting, together with a deployment framework showing that lightweight post-hoc recalibration is an effective safety intervention for African-language LLMs.

2 Related Work

Neural networks are often overconfident, and temperature scaling is a standard single-parameter post-hoc calibration method [1]; for free-form generation, sampling-based methods such as semantic entropy estimate uncertainty without labelled data [2]. Tomani et al. [3] show that uncertainty-based abstention improves correctness, reduces hallucinations, and increases safety, motivating uncertainty estimation as a deployment-time safety mechanism. African-language evaluation has advanced through human-translated benchmarks, including Uhura for truthfulness and scientific question answering [4] and AfriMMLU within IrokoBench for knowledge evaluation [5], alongside models such as AfroLlama [6]. However, prior work has focused primarily on capability and benchmark performance. To our knowledge, calibration in African languages, compared across models and paired with mitigation strategies, has not been systematically studied.

3 Methods

3.1 Models

We evaluate four open-weight models chosen to span size, multilinguality, and origin: a small instruction-tuned model, a larger one, a multilingual one, and an Africa-specific one (full specifications in Table 2, Appendix A). Open weights give direct log-probability access and full reproducibility without proprietary APIs.

3.2 Datasets

We use Uhura-TruthfulQA [4] (English and six African languages; truthfulness; a variable number of candidate answers per question) and AfriMMLU [5] (18 languages; knowledge; four options per question). Because their language sets and tasks differ, we analyse them separately and never average across them. "African" denotes all languages other than English and French.

3.3 Scoring

For both datasets we score each option by the length-normalised log-probability of its answer text (a cloze reading; Appendix C) and take the softmax over options as the model's confidence. We deliberately avoid letter-logit scoring (reading the next-token probability of "A"/"B"/"C"/"D"): it silently collapses instruction-tuned models, with Gemma-4 returning "A" for 75% of items and scoring below chance, whereas answer-text scoring collapsed for no model.

3.4 Metrics

We report expected calibration error (ECE, 15 bins), overconfidence (mean confidence minus accuracy), Brier score, error-detection AUROC, and AUARC, the area under the accuracy-rejection

Table 1: English versus African-language performance, by model and dataset. "en" is English and "afr" is the mean over African languages; ECE(TS) is pooled ECE after per-language temperature scaling; AUARC measures abstention quality (higher is better). Models are accurate and calibrated in English but lose both in African languages, and recalibration restores calibration.

Model	Dataset	Acc(en)	Acc(afr)	ECE(en)	ECE(afr)	ECE(TS)	AUARC
Qwen3-4B-Instruct	Uhura-TruthfulQA	0.38	0.32	0.07	0.17	**0.03**	0.36
Gemma-4-12B	Uhura-TruthfulQA	0.33	0.34	0.21	0.26	**0.03**	0.39
Aya-Expanse-8B	Uhura-TruthfulQA	0.36	0.31	0.08	0.20	**0.04**	0.35
AfroLlama-v1	Uhura-TruthfulQA	0.29	0.33	0.18	0.13	**0.04**	0.35
Qwen3-4B-Instruct	AfriMMLU	0.53	0.30	0.06	0.15	**0.02**	0.35
Gemma-4-12B	AfriMMLU	0.30	0.27	0.31	0.34	**0.07**	0.29
Aya-Expanse-8B	AfriMMLU	0.52	0.29	0.06	0.18	**0.02**	0.34
AfroLlama-v1	AfriMMLU	0.36	0.28	0.22	0.20	**0.03**	0.31

curve (definitions in Appendix B).

3.5 Interventions

We evaluate three: per-language temperature scaling, selective abstention by per-language confidence threshold, and 5-shot in-context prompting with examples held out from evaluation (Appendix D).

4 Results

Table 1 summarises every model on both datasets; the central finding is the gap between English and African-language calibration.

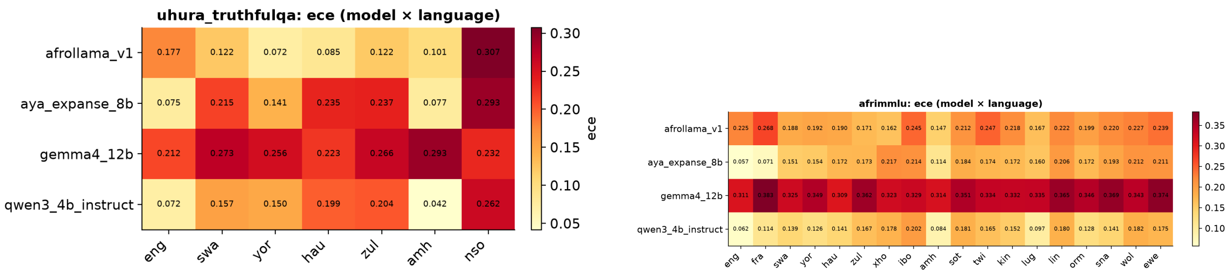
1. **Calibration degrades from English to African languages, consistently.** Across all four models and both tasks, ECE and overconfidence are two to three times higher in African languages (Table 1, Figure 1); models are most overconfident where they are least accurate.

2. **The accuracy collapse is a knowledge story.** On AfriMMLU the capable models fall from about 0.52 (English) to about 0.29 (African, near the 0.25 chance floor). On Uhura accuracy is uniformly low (about 0.31) with little English advantage, so Uhura supplies the calibration signal and AfriMMLU the accuracy decay.

3. **Mitigations are unequal.** Per-language temperature scaling cuts pooled ECE to about 0.02 to 0.07 for every model (Appendix Figure 3); abstention helps less, because uncertainty is a weak error-ranker (AUROC about 0.55), so AUARC sits only modestly above baseline. Few-shot prompting gives small gains for three models and backfires on Gemma (Appendix Table 3).

4. **Bigger is not safer; Africa-specific helps calibration, not knowledge.** Gemma-12B is the worst-calibrated model (ECE up to 0.34, overconfident across every topic), while AfroLlama is the only model better calibrated in African languages (Uhura ECE 0.18 to 0.13) yet no more accurate.

Findings are consistent across all four models and both datasets; per-language bootstrap confidence intervals are reported in the repository.



(a) Uhura-TruthfulQA (7 languages)

(b) AfriMMLU (18 languages)

Figure 1: Expected calibration error by model and language. ECE is low in English and French and rises across African languages for every model; Gemma is worst throughout.

5 Discussion and Limitations

Implications for African AI safety. In African languages these models are at once less accurate and more overconfident, the worst combination for trust, since confident wrong answers reach users precisely where independent verification is hardest. Encouragingly, much of the failure is an output-layer problem: cheap, post-hoc, per-language temperature scaling closes most of the calibration gap with no retraining, so safer deployment is feasible now. Abstention alone is insufficient because the uncertainty signal is weak, so the durable fix remains better African-language data and training. We distil this into a deployment framework (Appendix E, Figure 2): recalibrate, then route on calibrated confidence to answer, request more context, or abstain and defer, logging low-confidence queries to drive data collection.

Limitations. Accuracies are near chance on these hard, low-resource tasks, so we document systematic overconfidence rather than catastrophic confident-wrongness, and absolute ECE is moderate except for Gemma. The weak uncertainty signal (AUROC about 0.55) bounds how much abstention can help. We cover four open models and two benchmarks; reasoning and frontier closed models were out of scope. About 15% of Uhura items lack a recovered topic, so the category analysis uses the labelled subset.

Future work. Reasoning-model and closed-model references; in-context calibration at scale; the offline data-collection loop in Figure 2; and human-grounded evaluation of abstention.

6 Conclusion

Open LLMs are confidently wrong in African languages, less accurate and more overconfident than in English, uniformly across four models and two tasks. Post-hoc per-language temperature scaling is a cheap, effective safety lever that closes most of the calibration gap, and our deployment framework shows how to combine recalibration with thresholded abstention for safer real-world use.

Code and data. All code, predictions, figures, and tables are in the repository: `https://github.com/rexsimiloluwah/aims-team-apart-ai-safety-hackathon`. Datasets: `masakhane/uhura-truthfulqa` and `masakhane/afrimmlu`.

LLM usage. We used Claude to help build and debug the evaluation pipeline and to draft this report; all results were produced by our code and independently verified.

References

[1] Guo, C., Pleiss, G., Sun, Y., and Weinberger, K. Q. (2017). On Calibration of Modern Neural Networks. *Proceedings of the 34th International Conference on Machine Learning (ICML)*, PMLR 70:1321–1330. arXiv:1706.04599.

[2] Kuhn, L., Gal, Y., and Farquhar, S. (2023). Semantic Uncertainty: Linguistic Invariances for Uncertainty Estimation in Natural Language Generation. *International Conference on Learning Representations (ICLR)*. arXiv:2302.09664.

[3] Tomani, C., Chaudhuri, K., Evtimov, I., Cremers, D., and Ibrahim, M. (2024). Uncertainty-Based Abstention in LLMs Improves Safety and Reduces Hallucinations. arXiv:2404.10960.

[4] Bayes, E., Azime, I. A., Alabi, J. O., Kgomo, J., Eloundou, T., Proehl, E., Chen, K., Khadir, I., Etori, N. A., Muhammad, S. H., Mpanza, C., Thete, I. P., Klakow, D., and Adelani, D. I. (2024). Uhura: A Benchmark for Evaluating Scientific Question Answering and Truthfulness in Low-Resource African Languages. arXiv:2412.00948.

[5] Adelani, D. I., et al. (2024). IrokoBench: A New Benchmark for African Languages in the Age of Large Language Models. arXiv:2406.03368.

[6] Jacaranda Health (2024). AfroLlama-V1. Hugging Face. `https://huggingface.co/Jacaranda/AfroLlama_V1`.

A Models

Table 2: Open-weight models evaluated. [TEAM: verify AfroLlama parameter count.]

Model	**Params**	**Type**	**Hugging Face ID**
Qwen3-4B-Instruct	4B	Instruction-tuned	`Qwen/Qwen3-4B-Instruct-2507`
Gemma-4-12B	12B	Instruction-tuned	`google/gemma-4-12B-it`
Aya-Expanse-8B	8B	Multilingual	`CohereLabs/aya-expanse-8b`
AfroLlama-v1	8B	Africa-specific (LLaMA-based)	`Jacaranda/AfroLlama_V1`

B Metric definitions

For an item with options o and gold answer y, the cloze score of option o with tokens $o_1, \ldots, o_L$ given context x is the length-normalised log-probability

$$s(o) = \frac{1}{L} \sum_{t=1}^{L} \log p(o_t \mid x, o_{<t}),$$

$$p(o) = \mathrm{softmax}_o\, s(o), \qquad \hat{y} = \arg\max_o p(o), \qquad c = \max_o p(o).$$

Over N items with correctness $y_i \in \{0, 1\}$, confidence c_i, and $B = 15$ equal-width confidence bins $\{B_b\}$:

$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{|B_b|}{N} \left|\mathrm{Acc}(B_b) - \mathrm{conf}(B_b)\right|.$$

$$\mathrm{Overconfidence} = \frac{1}{N} \sum_i c_i - \frac{1}{N} \sum_i y_i.$$

$$\text{Brier} = \frac{1}{N}\sum_i (c_i - y_i)^2.$$

AUROC is the area under the ROC curve using c to rank correct against incorrect items. AUARC is the area under the accuracy-rejection curve: items are answered most-confident-first and accuracy on the answered subset is integrated over coverage. Temperature scaling rescales the logits by a single $T > 0$, $p_T(o) = \text{softmax}_o\big(s(o)/T\big)$; we fit $T^\star = \arg\min_T \text{ECE}$ on a seeded 50% calibration split and report ECE on the held-out half.

C Prompts

Each option is scored as a continuation of the prompt; the prompt lists no answer letters.

Uhura-TruthfulQA (truthfulness)

```
Q: {question}
A:
```

AfriMMLU (knowledge)

```
The following is a multiple-choice question about {subject}.  Give the correct
answer.
Question:  {question}
Answer:
```

For 5-shot prompting, the prompt is prefixed with five solved examples in the same language, each formatted as the template above followed by its gold answer text and separated by blank lines; the examples are held out from evaluation.

D Interventions

Temperature scaling. One temperature T per language, fit by minimising ECE on a seeded 50% split and evaluated on the other half, searching the grid

$$T \in \{0.5, 0.6, 0.7, 0.8, 0.9, 1.0, 1.2, 1.5, 2.0, 3.0, 5.0\}.$$

Abstention. The model answers only when confidence c exceeds a per-language threshold chosen from the risk-coverage curve at a target coverage; we report AUARC over all coverages. **Few-shot.** Five in-context examples per language, drawn from held-out items and excluded from evaluation.

E Deployment framework

F Additional results

Per-dataset figures referenced in the main text. Per-language bootstrap confidence intervals accompany each per-language table in the repository under `artifacts/_comparison/`.

Table 3: Zero-shot versus 5-shot on Uhura (values shown as zero-shot → 5-shot). Few-shot gives small gains for three models and backfires on Gemma.

Model	**Accuracy**	**ECE**	**Overconfidence**
Qwen3-4B-Instruct	$0.33 \rightarrow 0.36$	$0.16 \rightarrow 0.14$	$0.15 \rightarrow 0.13$
AfroLlama-v1	$0.32 \rightarrow 0.34$	$0.14 \rightarrow 0.12$	$0.14 \rightarrow 0.11$
Aya-Expanse-8B	$0.32 \rightarrow 0.33$	$0.18 \rightarrow 0.17$	$0.18 \rightarrow 0.16$
Gemma-4-12B	$0.34 \rightarrow 0.31$	$0.25 \rightarrow 0.33$	$0.25 \rightarrow 0.33$

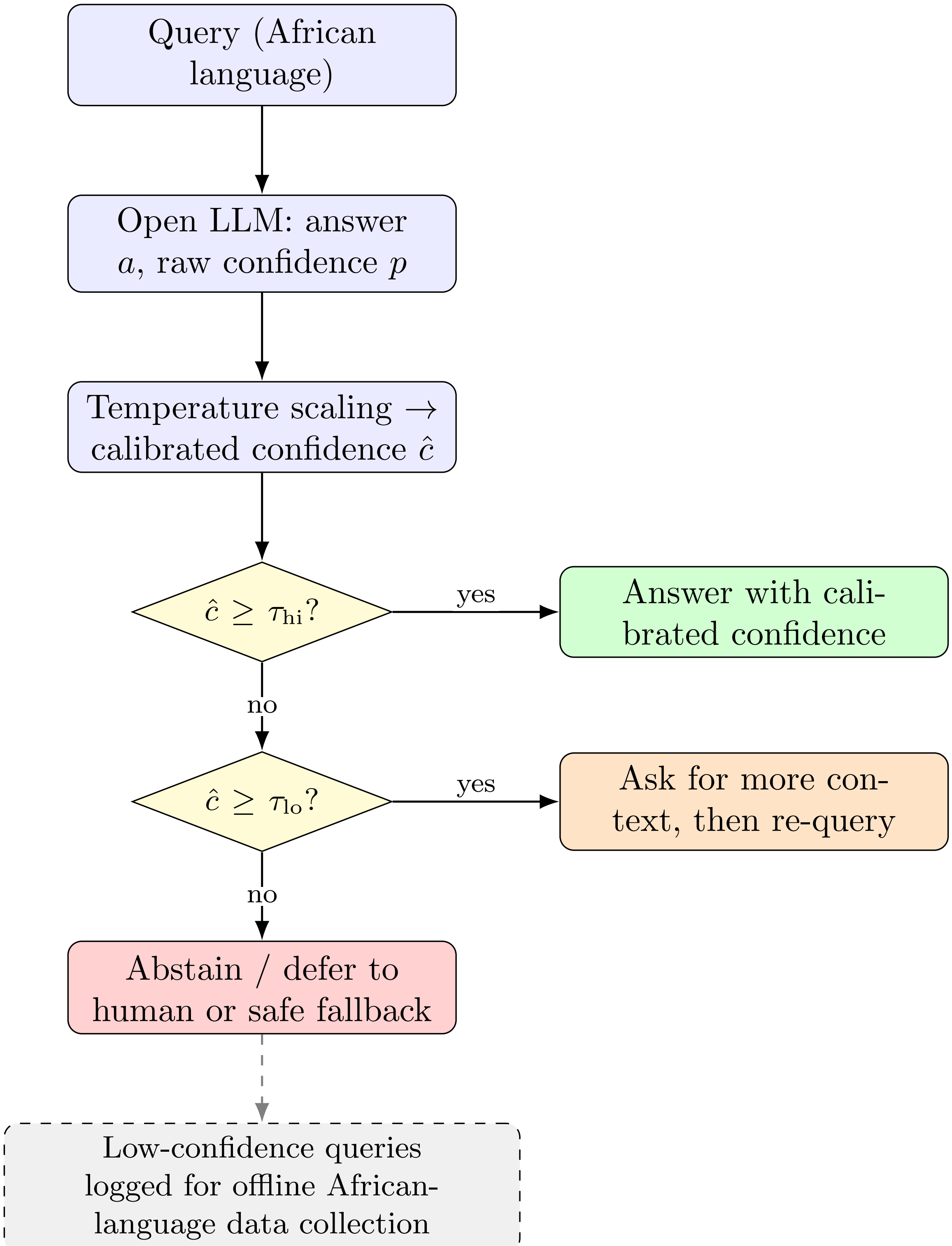
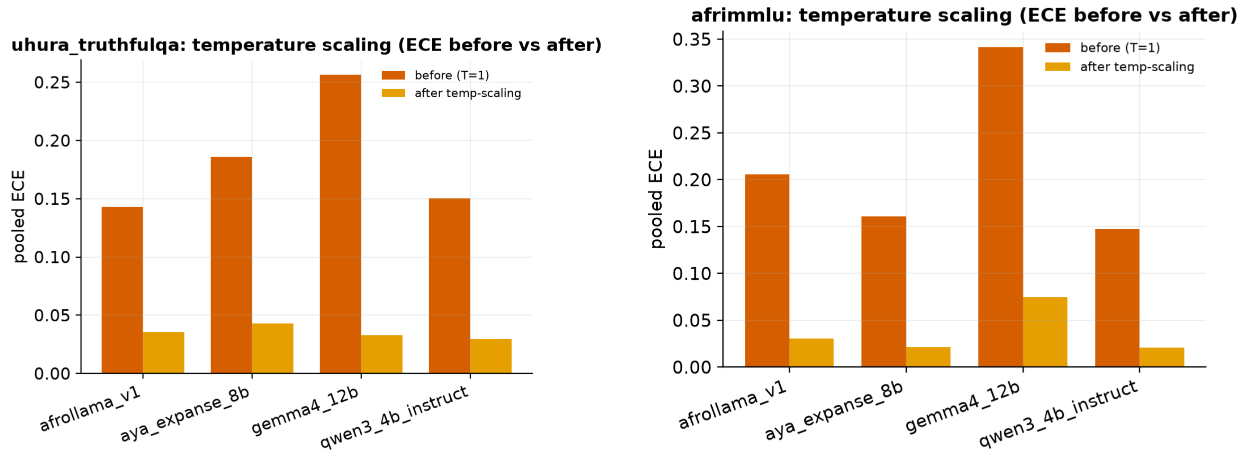
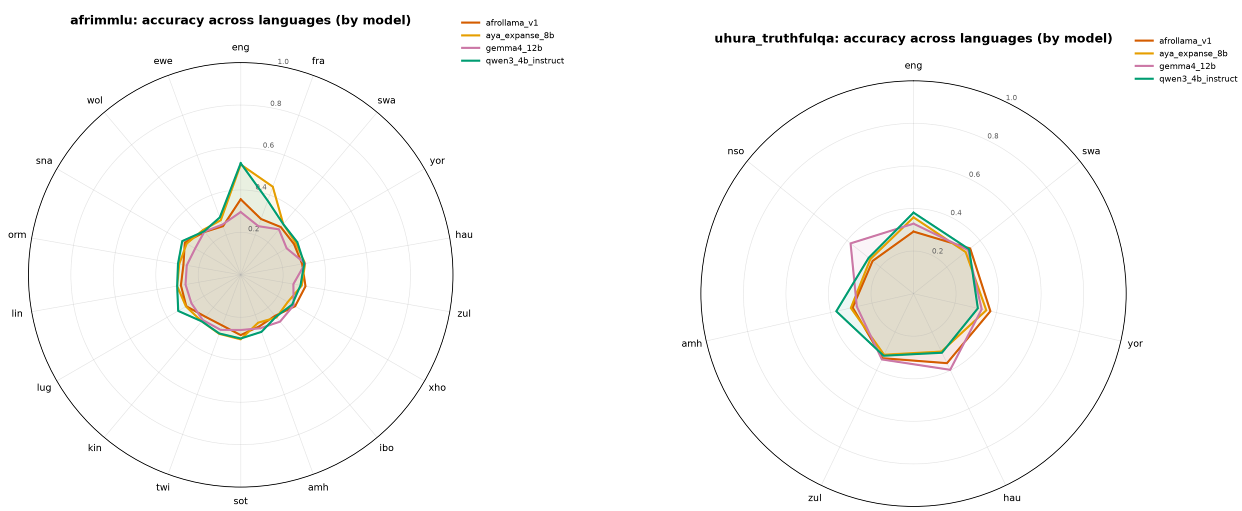


Figure 2: Deployment framework for LLMs in African languages. Recalibrate first, then route on the calibrated confidence $\hat{c}$: high values are answered, medium values trigger a request for more context, and low values are abstained or deferred. Thresholds τ are set per language from the risk-coverage curve.



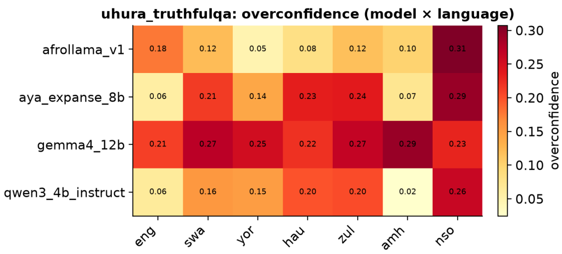
(a) Uhura-TruthfulQA **(b)** AfriMMLU

Figure 3: Temperature scaling: pooled ECE before versus after per-language recalibration; ECE drops to about 0.03 for all models.

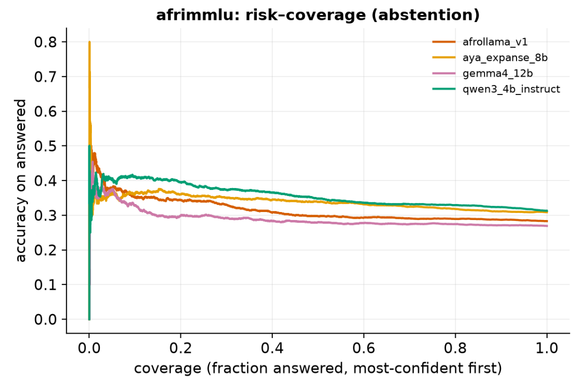


(a) AfriMMLU accuracy by language **(b)** Uhura accuracy by language

Figure 4: Accuracy across languages by model (factuality decay). Long English and French spokes collapse toward chance across African languages.

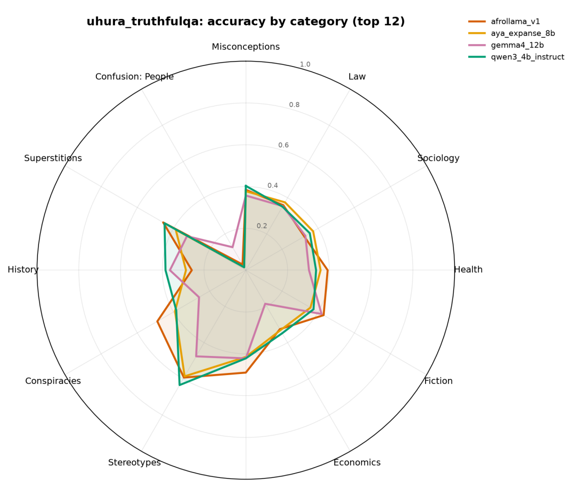


(a) Uhura overconfidence by language

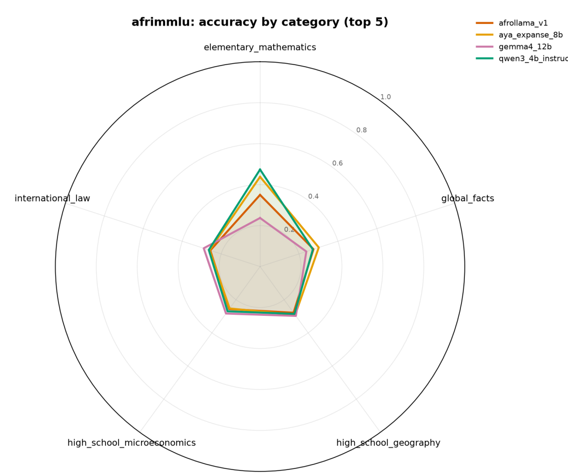


(b) AfriMMLU risk-coverage (abstention)

Figure 5: Overconfidence by language (left) and the accuracy-rejection trade-off under abstention (right).



(a) Uhura, by topic (top 12)



(b) AfriMMLU, by subject

Figure 6: Accuracy across question categories by model.